

JESSICA MARIANA JOHN

Sharjah, UAE

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Summary

Electronics and Communication Engineering student at BITS Pilani Dubai with a strong coding foundation built through full-stack web development. I enjoy combining hardware-level thinking with software execution, from database systems to AI-integrated applications and I'm currently expanding my skills through independent projects.

Education

Delhi Private School, Sharjah

April 2011 – March 2025

Subjects: Computer Science, Mathematics, Physics, Chemistry

Grade: 95/100

BITS Pilani, Dubai

September 2025 – Present

Bachelor of Engineering in Electronics and Communication

CGPA: 7.81/10

Experience

Institution of Engineers India (IEI) - BITS Pilani Dubai Chapter

July 2026 – Present

Electrical Co-Head

- Coordinate planning and execution of electrical-track events for the year, working alongside the head to align on scope, logistics, and timelines.
- Generate and pitch event ideas spanning both electrical and electronics engineering interests to engage chapter members.

Make A Difference (M.A.D) Club - BITS Pilani Dubai

July 2026 – Present

Volunteer Manager

- Manage volunteer coordination and attendance tracking for club events and initiatives.

Projects

Anonymous Letter Writing Web App | *React, Vite, Tailwind CSS, Supabase*

July 2026

- Built full-stack anonymous letter app; enforced anonymity at the DB layer via RLS and a security-definer function excluding author identity from public queries.
- Automated future-dated letter unlocking, triggering status changes and notifications server-side with zero client polling.

AI Flashcard Generator | *JavaScript, Express, Groq, Supabase*

July 2026

- Built AI flashcard generator parsing PDFs/DOCX into exact-count decks via the Groq API, shared across Express (local) and Vercel serverless (prod).
- Added gamified study system (XP, streaks, spaced review) with Supabase RLS-secured sync and session-authenticated AI endpoint.

Real-time Collaborative Notes | *Python, Flask, Socket.IO, SQLAlchemy, JavaScript, HTML, CSS*

June 2026

- Built real-time collaborative notes app with WebSocket rooms for simultaneous multi-user editing via Flask and Socket.IO.
- Implemented debounced live sync and auto-save; deployed on Render with Unicorn/Gevent for production WebSocket support.

Link Shortener | *Python, Flask, SQLite, SQLAlchemy, Chart.js, HTML, CSS*

June 2026

- Built full-stack URL shortener with unique code generation, click logging, and a REST API for analytics.
- Built Chart.js dashboard visualizing 7-day click trends and top referrers.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL

Developer Tools: VS Code, Git, Vercel, Render, Supabase, Figma, AutoCAD

Technologies/Frameworks: GitHub, Flask, MySQL, SQLAlchemy, SQLite, Chart.js, Node.js, Socket.IO, React, Express, Tailwind CSS, Vite, Framer Motion

Leadership / Extracurricular

High School Basketball Team

December 2023 – January 2025

Team Captain

Delhi Private School, Sharjah

- Captained the team to 3 tournament wins, encouraging consistent performance through structured preparation and on-court leadership.
- Received a Special Mention Award for remarkable individual performance in basketball, recognized among peers and faculty.